



conmed

Automated patient reconciliation

Reconciled before the visit.

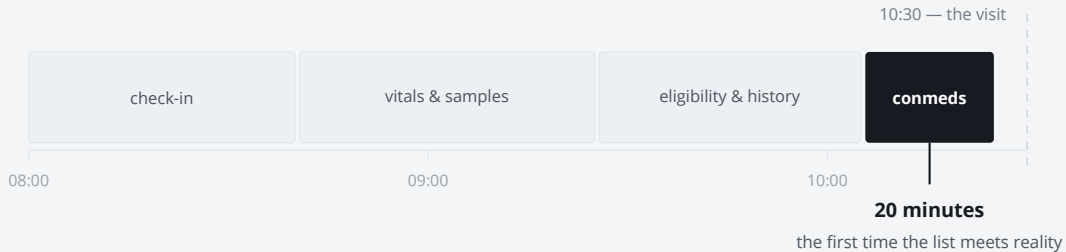
Pre-visit concomitant medication
reconciliation for clinical trial sites.

HackMIT 2026 · Regeneron Track



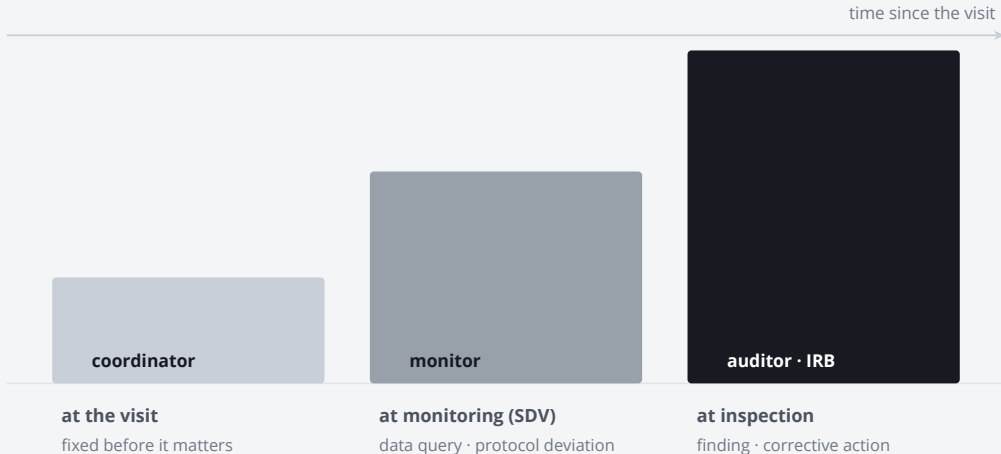
The medication list is stale, and the visit is now

Everything else is decided before the visit. The medication list is not.



Cheap to fix at the visit. Expensive to find later.

The same error, caught at three different points, by three different people.

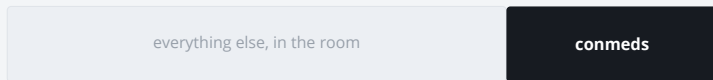


Ask days before, not minutes before

Move the conversation upstream and the coordinator starts from a finished draft.

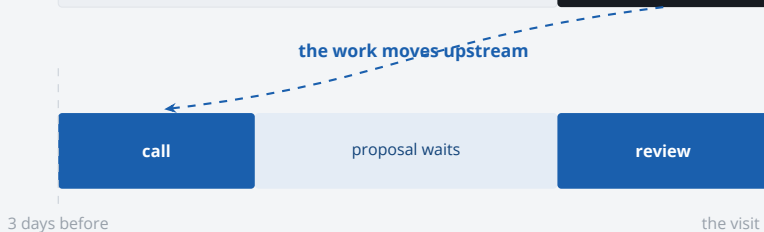
Today

at the visit



With conmed

three days earlier



Hours back, every visit round

The agent places the calls. The coordinator only reviews.



manual



30 calls × 12 min of coordinator time

4 h 30 back, per round

conmed



1 h 30

30 reviews × 3 min to approve a draft

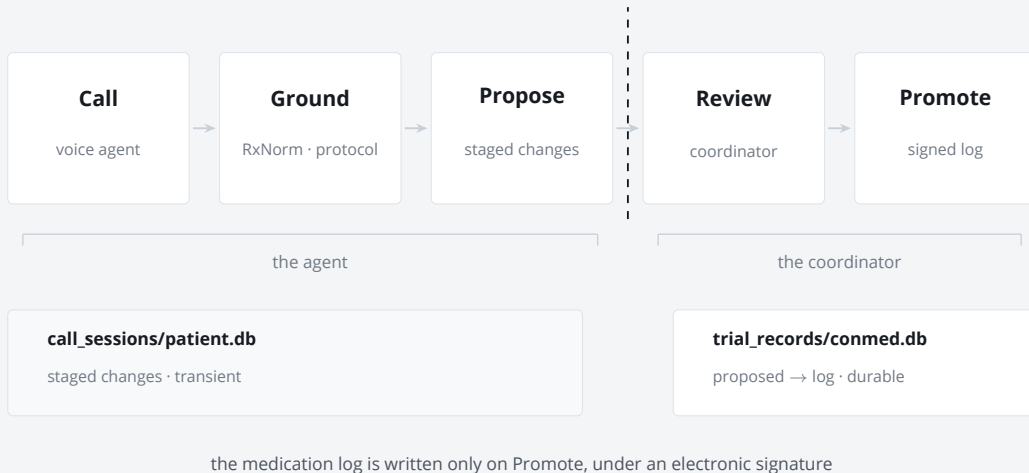
illustrative baseline: 30 participants, 12 min of manual reconciliation each, 3 min to review a finished draft.

The calls run in parallel. Only the exceptions reach a person.

The agent proposes. A coordinator decides.

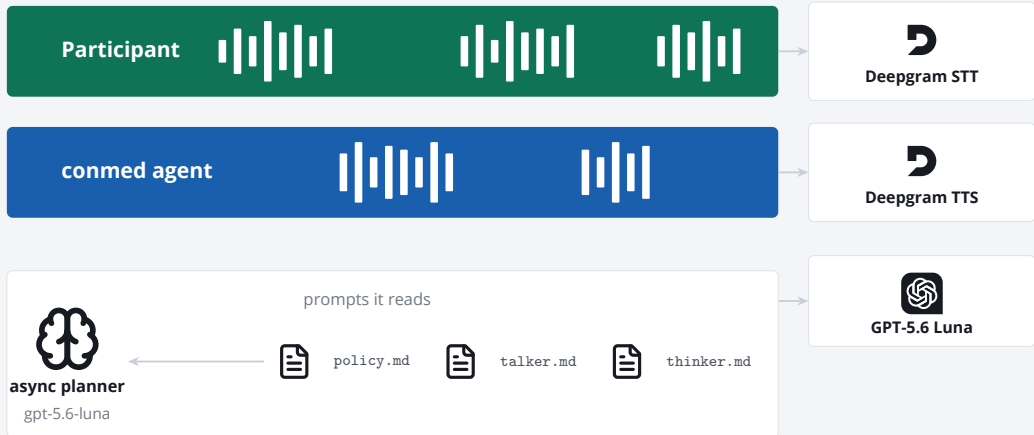
Nothing reaches the medication log without a signature.

no agent write



Turn by turn, and the agent never waits

Deepgram hears and speaks; GPT-5.6 Luna thinks. The prompts are files you can read.



Open source (MIT) · swap the model or the prompts

The model is not the source of truth

Left alone, it invents drugs — and invents words nobody said.

AP News

AI transcription tool used in hospitals invents things no one ever said

“hyperactivated antibiotics” — a drug that does not exist, invented in a patient’s notes.

26 Oct 2024

STAT

Watson recommended “unsafe and incorrect” cancer treatments

recommended a drug that could cause severe or fatal bleeding — for a patient already bleeding.

25 Jul 2018

Science

AI transcription tools “hallucinate”, too

made up whole sentences in about 1.4% of the transcriptions tested.

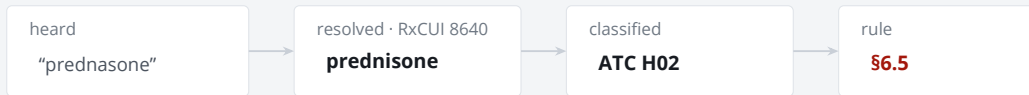
26 Apr 2024

Hallucination is a property to design around, not a bug to patch.

as reported by AP News, STAT and Science

The agent may not answer from memory

Drug facts come from RxNorm; patient facts come from the record. It is not allowed to guess.

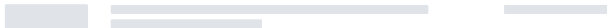


every hop is a lookup, not a guess

RxNorm / RxClass · NLM

Review queue

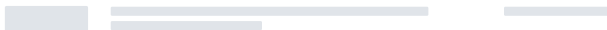
6 proposed changes



Prohibited

Prednisone
20 mg daily · oral

Protocol §6.5



● matched drug
prednisone

● matched class
systemic corticosteroid
ATC H02

● protocol section
§6.5
excluded above 10 mg/day

What is on file, and what the call proposed

Confirm, edit or reject one row at a time. The agent's reasoning is one click away.

Review queue

6 proposed changes · 4 need a decision

on file

from the call

Add

—



Atorvastatin 20 mg nightly



pending

Modify

Metformin 500 mg twice daily



Metformin 1000 mg twice daily



accepted

Stop

Ibuprofen 400 mg as needed



stopped — GI bleed reported



rejected

each row: confirm, edit, reject · and “why” opens the tool trace

Promote 1 accepted

The record a monitor will ask for

21 CFR Part 11: why it changed, who approved it, and when.

Audit trail

append-only

10:24	Voice agent reason: —	proposed stop ibuprofen
10:31	J. Okafor, CRC reason: GI bleed reported	accepted
10:31	J. Okafor, CRC reason: confirmed with participant	edited dose
10:32	J. Okafor, CRC reason: electronic signature applied	promoted 1 change

Electronic signature

J. Okafor, CRC

Clinical research coordinator

Approved

2026-09-19 10:32

A signature carries printed name,
timestamp and meaning (11.50).

Every entry carries who / when / why.

One machine, no network

Eight real Regeneron protocols and 42 participants — all synthetic at site level.

Everything local



conmed.db

the trial record



patient.db

the call in progress



drugs.db

RxNorm



general_health.db

plain-language guidance

delete any of them and rebuild from the source data

Or a server

`DATABASE_URL=`

`postgres://...`

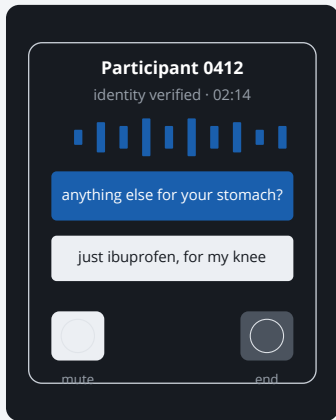
no database is opened by path:
one environment variable, same code

MIT · open source

the same schema, three methods

See it make the call

A call three days before the visit — heard, understood, staged.



scan — the whole project

github.com/Samuel-O-M/HackMIT-2026

What it changes

The quiet majority is the point: twenty-seven medications confirmed without anyone being asked.

30

calls run

69

changes proposed

27

confirmed unchanged

8

real protocols

coordinator

- starts from a draft, not a blank form

sponsor

- fewer deviations found late

participant

- nothing to remember or repeat

Open source (MIT) · point it at your own model · pilot it at one site



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Thank you · questions

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